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Research article

A macroalgae-based biotechnology for water remediation: Simultaneous removal of Cd, Pb and Hg by living *Ulva lactuca*

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ABSTRACT

Metal uptake from contaminated waters by living *Ulva lactuca* was studied during 6 days, under different relevant contamination scenarios. In mono-metallic solutions, with concentrations ranging from 10 to 100 μg L^{-1} for Hg, 10–200 μg L^{-1} for Cd, and 50–1000 μg L^{-1} for Pb, macroalgae (500 mg L^{-1}, d.w.) were able to remove, in most cases 93–99% of metal, allowing to achieve water quality criteria regarding both surface and drinking waters. In multi-metallic solutions, comprising simultaneously the three metals, living macroalgae still performed well, with Hg removal (c.a. 99%) not being significantly affected by the presence of Cd and Pb, even when those metals were in higher concentrations. Removal efficiencies for Cd and Pb varied between 57 and 96%, and 34–97%, respectively, revealing an affinity of *U. lactuca* toward metals: Hg > Cd > Pb. Chemical quantification in macroalgae, after bioaccumulation assays demonstrated that all Cd and Hg removed from solution was really bound in macroalgae biomass, while only half of Pb showed to be sorbed on the biomass. Overall, *U. lactuca* accumulated up to 209 μg g^{-1} of Hg, up to 347 μg g^{-1} of Cd and up to 1641 μg g^{-1} of Pb, which correspond to bioconcentration factors ranging from 500 to 2200, in a dose-dependent accumulation. Pseudo-first order, pseudo-second order and Elovich models showed a good performance in describing the kinetics of bioaccumulation, in the whole period of time. In the range of experimental conditions used, no mortality was observed and *U. lactuca* relative growth rate was not significantly affected by the presence of metals. Results represent an important contribution for developing a macroalgae-based biotechnology, applied for contaminated saline water remediation, more "green" and cost-effective than conventional treatment methods.

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1. Introduction

Marine algae are responsible for over 50% of the O_2 produced globally (Chojnacka, 2009), having long been used for food, agricultural, cosmetic and pharmaceutical purposes (Caliceti et al., 2002; He and Chen, 2014; Kumar et al., 2013). Recently, there is a strong enthusiasm involving them in the pursuit of "greener" technologies and environment protection (Sekabira et al., 2011; Torres et al., 2008).

It is well recognized that water availability can be compromised by human activities, such as mining, smelting, or coal burning, which mobilize and release large amounts of metals into the environment (Freitas et al., 2008). Given the awareness for the harmful effects that potential toxic elements such as Cd, Pb and Hg – the so-called toxic trio – may have on human and animal health, environmental policies are becoming more stringent every day.

Conventional water treatment procedures present major drawbacks (Farooq et al., 2010), and fail or become prohibitively expensive in reducing metal levels to concentrations such low as required by law (He and Chen, 2014). Due to their large abundance worldwide and ability to bind and retain chemical elements from the surrounding environment (Jayakumar et al., 2014; Kamala-

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Kannan et al., 2008) macroalgae may be the needed alternative, forming the basis of new remediation biotechnologies, more efficient and cost-effective.

In the last years, an intense investigation has focused on the application of dried or pre-treated algae biomass to promote the removal of metals from aqueous solutions (biosorption) (Romera et al., 2007; Trinelli et al., 2013; Wang and Chen, 2009), however, the use of living organisms (bioaccumulation) (Chojnacka, 2010), although less studied, may be advantageous in some scenarios. While growing, living macroalgae can remove simultaneously metals and nutrients (nitrates and phosphates) from aquaculture wastewaters and capture CO_2 emissions (Henriques et al., 2015; Nielsen et al., 2012). Additionally, living cells may lead to lower residuals levels of contaminant in solution due to additional intracellular accumulation (Chojnacka, 2007). As a treatment method, bioaccumulation will require simpler installation when compared with biosorption (Chojnacka, 2010), and avoid the need for additional processes, such as drying, processing and activation prior to use (Li et al., 2013), which are indispensable for biosorption.

A significant part of the laboratory studies concerning accumulation of metals by living macroalgae (Lee and Wang, 2001; Masakorala et al., 2008) reports short exposure times (Turner et al., 2008a; Wang and Dei, 1999) and ignores the mathematical description of the process kinetics. Among the metals studied less attention is given to Hg (Costa et al., 2011), and above all no study dealt with the effects of simultaneous presence of metals (mixtures) on macroalgae growth rate (Han and Choi, 2005; Han et al., 2008; Turner et al., 2008a; Vasconcelos and Leal, 2001). In fact, most of researchers follow an idealistic approach, assessing the performance of the removal process for mono-metallic solutions (Chojnacka, 2009). However, real waters are complex systems, containing a multiplicity of different ions, which interact and compete for binding sites, affecting removal efficiency. Moreover, extremely high concentrations of metal are often used, which does not comply with the values found in real contaminated waters bodies (Lopes et al., 2014; Rocha et al., 2013). Besides the scarcity of works conducted for multi-metallic systems, literature also lacks on studies performed in saline water.

Ulva lactuca is a green marine macroalga (*Chlorophyta*), very common at coastal areas, which owns several attractive characteristics for water remediation. It is perfectly adapted to the salinity variations occurring in estuaries due to tides (Costa et al., 2011; Yamashita et al., 2009) and it withstands moderate eutrophic conditions. Its structure is relatively simple, with a just 2 cell layers thick and sheet-like thallus, which translates into a large surface area, containing structurally uniform and physiologically active cells (Sari and Tuzen, 2008). The cell wall is constituted by cellulose along with a high percentage of proteins bonded to polysaccharides (Romera et al., 2007; Trinelli et al., 2013), which comprises several functional groups such as amino, hydroxyl, carboxyl, and sulfate (Romera et al., 2007). Additionally, this alga grows fast (Nikolaisen and Jensen, 2013), since their photosynthetic products are quickly converted to cell growth.

Thus, the purpose of this work was to study and evaluate the accumulation of metals by *Ulva lactuca* from saline water. The efficiency of the removal process carried by the macroalga was assessed for: i) mono-metallic solutions of Hg, Cd and Pb, in a range of different environmentally relevant concentrations; ii) multi-metallic solutions containing Hg, Cd and Pb in equal concentration, and at different concentrations. Evaluation of the influence of the chosen metals on *Ulva* growth rate, during accumulation, in both mono and multi-system was also performed.

2. Materials and methods

2.1. Material and chemicals

All chemicals used in this work were of analytical grade, and were used as received from the suppliers. Nitric acid 65% (Suprapur®) and certified stock solutions of mercury (1001 ± 2 mg L^{-1}), cadmium (1000 ± 2 mg L^{-1}) and lead (1000 ± 2 mg L^{-1}) nitrate were purchased from Merck. All the material used in the assays was previously washed in Derquim 5%, rinsed in Milli-Q water (18 MΩ cm^{-1}), soaked in 25% HNO_3 for at least 24 h and subsequently rinsed with Milli-Q water.

2.2. Macroalgae collection and maintenance

Ulva lactuca collection was held in the Mondego estuary (Portugal, 40°08′N, 8°50′W). Seaweed were transported to the laboratory in isothermal plastic bags, containing local water, and subsequently rinsed with seawater to remove extraneous matter, such as debris and epibionts. A macroalgae sample was immediately freeze dried for further quantification of the inherent metal concentrations (baseline). Until the beginning of the experiments, macroalgae were maintained in 30 L clear glass aquariums equipped with air pump and filled with filtered seawater enriched with Provasoli stock solution (Costa et al., 2011), under natural light, and at room temperature of 20 ± 2 °C.

Dry/fresh weight ratio of *U. lactuca* was assessed by gravimetric analysis. Several pieces were cut from the macroalgae thallus, weighted and dried at 40 °C until constant weight.

Seawater for experiments and macroalgae maintenance was collected at Vagueira beach (Portugal, 40°33′N, 8°46′W), filtered through 0.45 μm Millipore membrane and stored in the dark at 4 °C until further use. Seawater pH, conductivity, salinity, and major and minor elements were measured, with values being in line with those recently reported in Henriques et al. (2015). Concentrations of Cd and Pb were below 2 and 4 μg L^{-1}, respectively, while total Hg concentration was 3.1 ng L^{-1} (Henriques et al., 2015).

2.3. Metal bioaccumulation experiments

U. lactuca was exposed, during 6 days, to mono- and multi-metallic solutions of Cd, Pb and Hg, according to the experimental conditions presented in Table 1.

Concentrations were selected intending to be environmentally relevant, i.e., 50, 200 and 1000 μg L^{-1} are, respectively the "old" limits for Hg, Cd and Pb in wastewaters discharges (Decree-Law No. 236/98, 1998; Directive 83/513/EEC, 1983; Directive 84/156/EEC, 1984). It should be remembered that European Commission has imposed the cessation of Cd and Hg emissions, losses and discharges, considering new stringent environmental quality standards (EQS) for surface waters (Directive 2013/39/EU, 2013).

Table 1
Experimental conditions of the metal bioaccumulation experiments performed.

Matrix	System	Metal		Concentration (μg L^{-1})
Spiked natural seawater	mono-metallic	Cd		10, 50, 100 and 200
		Pb		50, 100, 200 and 1000
		Hg		10, 50 and 100
	multi-metallic	M1	Cd	50
			Pb	50
			Hg	50
		M2	Cd	200
			Pb	1000
			Hg	50

Concentrations below and above the reference values used may represent, respectively, the levels in the aquatic ecosystem affected by wastewater discharges and the levels of untreated effluents.

Bioaccumulation assays were conducted in Schott Duran® glass bottles of 1 L, containing natural filtered seawater, spiked with desired metal(s). All the solutions were left to stabilize during 24 h before the beginning of the experiments. Small discs with uniform size (10 discs of 18 mm and 5 discs of 56 mm) were cut from the healthy thallus of *U. lactuca* (Wang and Dei, 1999), in a total of approximately 1.6 g L^{-1} fresh weight, and acclimatized in clean seawater for several hours prior to metal exposure. Metal uptake by living macroalgae along time was followed by determining the concentration of each metal in solution samples (5–10 mL) collected at defined crescent periods of time. All samples were acidified to pH ≤ 2 using Suprapur® HNO_3 and stored at 4 °C until further analysis. All bioaccumulation trials were performed in triplicate, under natural light, at room temperature of 20 ± 2 °C. Blanks (macroalgae in clean seawater) and controls (metal spiked seawater in the absence of algae) were always running in parallel with the experiments. No pH buffer was added to the system, and no pH adjustments were made at any time, since this could interfere with the metals uptake and cellular metabolism of the organisms (Henriques et al., 2015). However, pH was daily monitored. At the end of the experiments, macroalgae were harvested, rinsed and freeze dried for later quantification of metal content.

2.4. Metal quantifications

2.4.1. Metal concentration in solutions

Mercury quantification in water samples was performed by cold vapour atomic fluorescence spectroscopy (CV-AFS), on a PSA 10.025 Millennium Merlin Hg analyser and using $SnCl_2$ (2% m/v in HCl 10% v/v) as reducing agent. In the range of operational conditions used, quantification limit was 0.021 μg L^{-1} (n = 20), with an acceptable relative standard deviation among replicates <5%. Cadmium and Pb quantifications were performed by inductively coupled plasma mass spectrometry (ICP-MS), on a Thermo ICP-MS XSeries equipped with a Burgener nebulizer. The limits of quantification were 0.1 μg L^{-1} and 0.2 μg L^{-1} for Cd and Pb, respectively, with an acceptable relative standard deviation among replicates <10%. In order to avoid matrix interferences on metal quantification by ICP-MS due to seawater high salinity, all samples were diluted 20-fold prior analysis (in HNO_3 2% v/v).

2.4.2. Metal concentration in algal biomass

Total Hg concentration in macroalgae biomass was measured by thermal decomposition atomic absorption spectrometry with gold amalgamation using a LECO© AMA-254 (Henriques et al., 2015). Analysis was performed directly in the sample (1–20 mg, dry weight) without digestion or pre-specific treatment. At least three replicate measurements were carried out for each sample, and a maximum coefficient of variation of 10% was adopted as acceptance criteria. Quantification limit was 0.03 ng Hg (n = 20; 99.5% confidence level). Several blanks (i.e. an empty sample nickel boat) were run before and between sample analyses to ensure that mercury was not being accumulated between samples. The quality of the results was assessed by using Certified Reference Material (CRM) BCR-60 (*Lagarosiphon major*). All percentages of recovery for total Hg were within the range of 85–107% (n = 17).

For the determination of Cd and Pb contents in macroalgae, samples and certified reference material (BCR60) were previously digested, in duplicate (Monterroso et al., 2003). Briefly, about 200 mg of freeze-dried sample was accurately weighted into an acid-washed Teflon reactor; 4 mL HNO_3 (conc.) was added, and reactors were placed in an oven at 60 °C for 12 h, and then at 100 °C for 1 h. Afterward, 1 mL H_2O_2 was added, and reactors were heated at 80 °C for 1 h. After digestion, samples were diluted with Milli-Q water, filtered, and then analysed by ICP-MS. Percentages of recovery for Cd and Pb were, respectively, 85 and 92% (n = 7).

2.5. Evaluation of the effect of metal exposure on macroalgae growth

The influence of studied contaminants on *U. lactuca* growth, along with time, was evaluated by means of relative growth rate (RGR), calculated by the following equation:

$$RGR\left(\%\, day^{-1}\right) = ((\ln A_t - \ln A_0)/t) \times 100 \quad (1)$$

where A_0 is the initial algae area and A_t is the algae area at time t (Han et al., 2008).

Two algal discs were daily collected from each reaction bottle, placed in a transparent plastic sheet, and then scanned with a resolution of 200 ppi. Afterwards, discs area was determined using the software *ImageJ*. Overall, the combined mass of all discs collected along the experiment for RGR measurements is <10% of the initial total algal biomass.

2.6. Experimental data analysis

Metal uptake by macroalgae, defined as the amount of metal bound by unit of mass at a given time t (q_t, μg g^{-1}), was deduced from the mass balance between the initial metal concentration in the solution (C_0, μg L^{-1}) and the concentration after a particular period of contact time t (C_t, μg L^{-1}):

$$q_t = \frac{(C_0 - C_t)V}{m} \quad (2)$$

where V (L) is the volume of the solution and m (g) is the macroalgae biomass, in dry weight. If equilibrium is attained, $t = t_e$, $q_t = q_e$ and $C_t = C_e$ (residual metal concentration in solution). The percentage of metal removed from solution at time t (R_t, %) was obtained from:

$$R_t = \frac{(C_0 - C_t)}{C_0} \times 100 \quad (3)$$

2.6.1. Metal bioaccumulation kinetics

Bioaccumulation kinetics was studied through the most commonly used kinetic reaction models in sorption batch experiments: Lagergren pseudo-first-order model, Ho's pseudo-second order model and Elovich model (Ho et al., 2000). Their non-linear equations are presented below.

$$Lagergren\ pseudo-first-order\ model:\ q_t = q_e\left(1 - e^{-k_1 t}\right) \quad (4)$$

$$Ho's\ pseudo-second\ order\ model:\ q_t = \frac{q_e^2 k_2 t}{1 + q_e k_2 t} \quad (5)$$

$$Elovich\ model:\ q_t = \frac{1}{\beta}\ln(1 + \alpha\beta t) \quad (6)$$

where k_1 and k_2 are the rate constants of pseudo-first order (h^{-1}) and pseudo-second order (g μmol^{-1} h^{-1}), respectively; α represents the initial sorption rate (μmol g^{-1} h^{-1}) while β is a desorption

constant (g μmol^{-1}) (Henriques et al., 2015).

2.6.2. Statistical analysis of the results

GraphPad Prism was used to perform all statistical analysis in this work, as well as to plot all graphs and calculate all kinetic parameters. The goodness of fit was evaluated by analyzing coefficient of determination (R^2) and standard deviation of residuals ($S_{y.x}$) values. The relative error (E_r) between experimental and predicted values of q_e was also analysed (Henriques et al., 2015).

Akaike's Information Criterion (AIC) (El-Khaiary and Malash, 2011) was used to compare models, determining which model is more likely to be correct and quantifying how much more likely. For small sample size, AIC is calculated from the equation:

$$AIC = N \ln\left(\frac{SSE}{N}\right) + 2N_p + \frac{2N_p(N_p+1)}{N-N_p-1} \quad (7)$$

where SSE is the sum-of-squared deviations, N is the number of data points and N_p is the number of parameters in the model.

AIC values can be compared using Evidence ratio which is defined by:

$$\text{Evidence ratio} = \frac{1}{e^{-0.5\Delta}} \quad (8)$$

where Δ is the absolute value of the difference in AIC between the two compared models.

3. Results and discussion

3.1. Ulva lactuca characterization

The dry/fresh weight ratio of *U. lactuca* ranged between 0.21 and 0.33, corresponding to an average water content of 74± 4% (n = 20). These values are in line with those reported by Bruhn et al. (2011) for *U. lactuca* from the Danish coast (76.7 ± 2.7% of water content), and slightly lower than those found at New Zealand (79.6 ± 0.43% of water content) by Lamare and Wing (2001).

Natural concentrations of metal in macroalgal tissues were globally low, namely, 0.015 ± 0.004 μg g^{-1} for Cd, 0.300 ± 0.025 μg g^{-1} for Pb and 0.029 ± 0.003 μg g^{-1} for Hg (Eisler, 2010).

FTIR and SEM-EDS analysis on *Ulva lactuca* was previously done and results are published in Henriques et al. (2015).

3.2. Water bioremediation in mono-metallic solutions

The results of controls (dashed line - Fig. 1A–D) reveal that levels of Cd in solution, in the absence of macroalgae, remained stable throughout the full period of time in which the assays were held. By contrary, the kinetic profiles of all solutions in contact with *U. lactuca* were characterized by a rapid and marked decrease in C_t/C_0, at the initial period of time, representing a quick biosorption/bioaccumulation of metal by the living macroalgae. After just 12 h of contact time with 500 mg (d.w.) of macroalgae, levels of Cd in seawater were reduced by about 77.8, 78.4, 66.7%, respectively for initial concentrations of 10, 50, 100 μg L^{-1}, whereas for 200 μg L^{-1} a removal percentage of 83.8% was achieved after a period of time of 24 h. Results suggest that for higher concentrations a longer period of time is required in order to obtain similar removal efficiencies (in terms of % of metal removed). However, it should be noted that an increase in the initial concentration of Cd always leads to higher amounts of metal accumulated by living macroalgae - dose-dependent accumulation (e.g. 15.8, 66.4, 114 and 154 μg g^{-1} for 10, 50, 10 and 200 μg L^{-1}, respectively, after 12 h of contact time). Additionally, overall, initial sorption rate, v_0, also increases with the rise of initial concentrations of Cd (Table S1 in supplementary information), which can be attributed to the greater driving force allowing to overcome all mass transfer resistance of Cd between the liquid and macroalgae (Rocha et al., 2013).

All samples collected after 12 h for 10 μg L^{-1}, 24 h for 50 μg L^{-1} and 96 h for 100 μg L^{-1} (Fig. 1A–C) had concentrations of Cd below the "limit of quantification" of the method, underscoring the potential application of this seaweed in water bioremediation processes. Additionally, the kinetic pattern observed for 10 and 50 μg L^{-1} suggests that removal continued after the referred times, possibly until an equilibrium was established, with residual concentrations in solution near to zero.

The fittings of pseudo-first order (PFO), pseudo-second order (PSO) and Elovich models to experimental data regarding the bioaccumulation of Cd by *U. lactuca* over time are presented in Fig. 2, while respective estimated parameters and goodness of fit are summarized in Table S1 (supplementary material). Kinetic analysis was performed on a molar basis, in order to enable further comparison between metals.

All mathematical models studied adequately fit the experimental results over the entire course of experiments, which is corroborated by the values of the coefficient of correlation ($0.939 < R^2 < 0.993$). Generally, the best performance was attributed to PFO model (Table S1), which in fact described precisely qt over the full range of time, especially for 100 and 200 μg L^{-1}, where both PSO and Elovich models overestimated $q_{t,144h}$ (Fig. 2C–D).

The notable affinity of *U. lactuca* toward Cd was previously reported by Wang and Dei (1999), although results found by the authors suggested a slower accumulation than that observed in the present study. After 47 h of contact time, for 10 μg L^{-1}, Wang and Dei (1999) recorded values of Cd internalized by macroalgae between 15 and 20 μg g^{-1} (d.w.), while in our case, for an equal initial concentration of Cd, an accumulation of about 18 μg g^{-1} (d.w.) was achieved just in 12 h. Furthermore, authors always observed a linear relationship between q_t and t, without any steady-state being reached, which may due to the fact that exposure to metal only lasted 2 days (Wang and Dei, 1999).

Results for Pb (Fig. 3A–D) revealed that, as for Cd, seaweed are the main responsible for the decrease in concentrations of Pb in seawater. Yet, a slightly different behaviour from Cd was observed, particularly in the initial removal period. The initial decrease on C_t/C_0 for Pb was less pronounced, which is corroborated by the minor values of initial bioaccumulation rate, v_0 (Tables S1 and S2). In addition, after the quick initial sorption, the drop in C_t/C_0 becomes less marked with the rise of initial concentration of Pb in solution.

Nevertheless, experimental data demonstrated that living *U. lactuca* also has a remarkable capacity to remove Pb from saline waters. After 12 h levels of Pb in seawater were reduced by 49.5, 48.5, 34.4 and 21.2% for C_0, Pb of 50, 100, 200 e 1000 μg L^{-1}, whereas increasing the contact time to 48 h allowed to obtain reductions of 75.6, 78.8, 66.2 and 37.0%, respectively. As mentioned previously for Cd, the rise in initial metal concentrations, which represents an increase of the driving force, leads to higher initial sorption rates (Table S2) and larger amounts of Pb bioaccumulated.

For initial concentrations of 50, 100 and 200 μg L^{-1}, equilibrium was reached after 96 h (Fig. 3A–C), whereas the kinetic pattern observed for 1000 μg L^{-1} indicates that a longer period of exposure time is required in order to attain a steady-state (Fig. 3D).

Overall, there is a good agreement between experimental data and the fittings accomplished by the three kinetic models (Fig. 4A–D), with R^2 varying between 0.935 and 0.999, for 50–200 μg L^{-1} (Table S2). A worst performance was recorded for 1000 μg L^{-1}, in which case equilibrium has not reached, with coefficients of correlation ranging from 0.886 to 0.939.

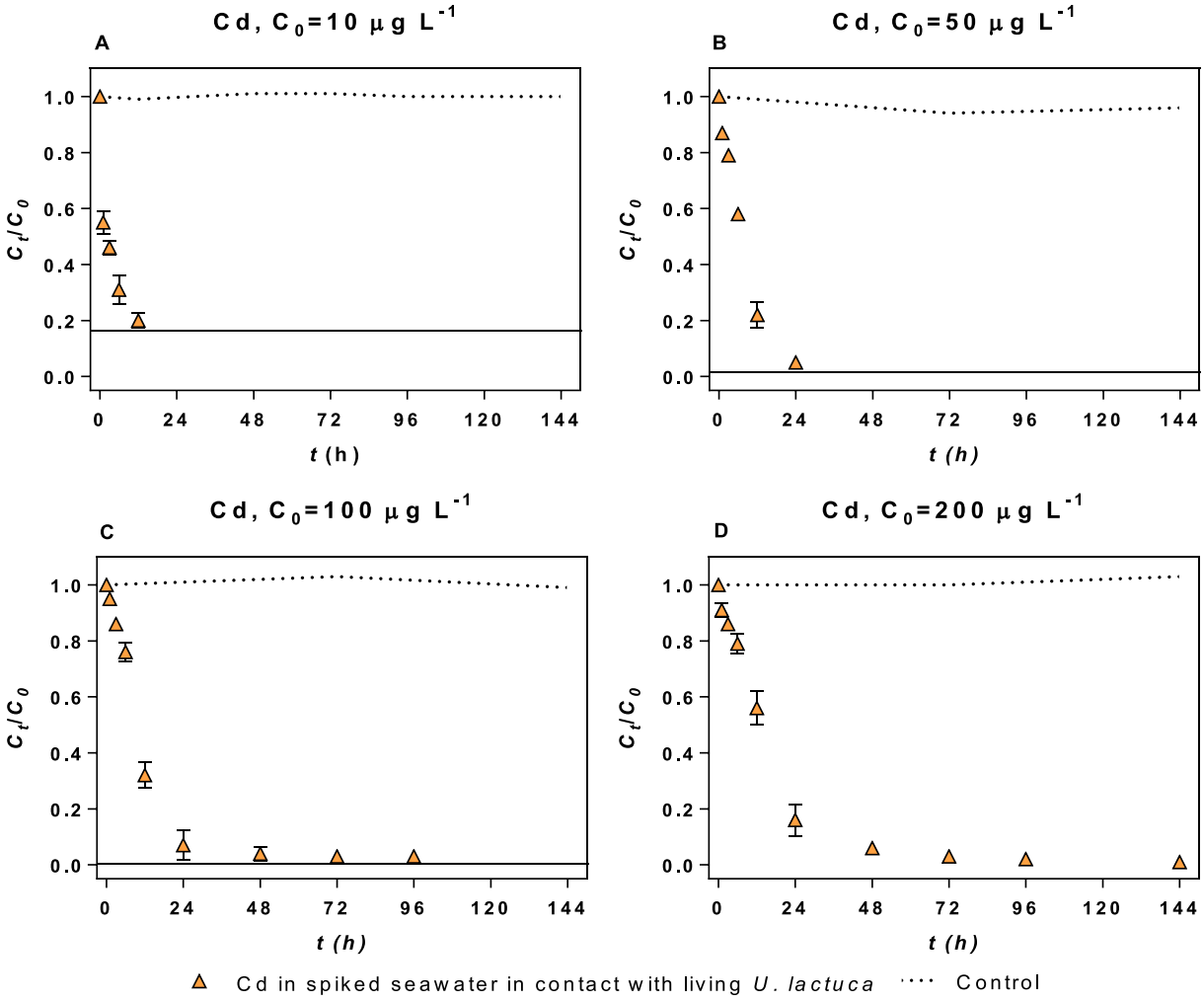


Fig. 1. Normalized concentrations (C_t/C_0) of Cd in spiked seawater along time (t, h) – mono-metallic solutions. The results are expressed as the mean ± standard deviation (n = 3). The solid line represents the value of the limit of quantification (in C_t/C_0).

Comparison between models by using AIC indicates that, in most cases, Elovich model is likely to be the most suitable to describe the bioaccumulation kinetics of Pb by the green macroalgae, which points to a heterogeneous uptake/bioaccumulation mechanism (Cheung et al., 2000). Moreover, as *qt vs t* for Cd and Pb followed different kinetics (PFO vs Elovich, respectively), probably the mechanisms behind the removal of these metals by the seaweeds will not be exactly the same. In their comparative study on biosorption of Cd, Zn, and Pb from aqueous solutions by dried marine macroalgae, Freitas et al. (2008) also found that PFO was better for the system Cd/*B. bifurcate*, whereas for Pb/*B. bifurcate* PSO performed better. Differences in behaviour were justified by the different properties of metals ions in solution and different interactions between metals and biomass (Freitas et al., 2008).

Despite the best performance in mathematics description of the kinetics over the entire period of time of trials, Elovich model did not estimated accurately the values of $q_{t,144h}$ (Fig. 4A–D). Amounts of Pb in macroalgae tissues predicted by the PSO for t = 144 h are closer to the experimental points than those under- and over-estimated by PFO and Elovich models, respectively.

For Hg (Fig. 5A–C), regardless the initial metal content in seawater, kinetic profiles are characterized by a fast and noticeable decrease in C_t/C_0, in the initial period of time, followed by a subtle decrease in C_t/C_0, until equilibrium was established (at 72 h for 10 µg L^{-1} and at 96 h for 50 and 100 µg L^{-1}). Overall, macroalgae-free experiments show a constant metal concentration with time, proving that removal of Hg may only be attributed to interactions metal-*Ulva lactuca*.

After 12 h of contact time, green macroalgae led to removal percentages of 59.7, 49.3 and 41.2%, for 10, 50, and 100 µg L^{-1}, respectively, corresponding to Hg contents in macroalgal tissues of 13.9, 51.4 and 86.7 µg g^{-1}, respectively. Like it was verified for Cd and Pb, uptake of Hg is dose-dependent, being in accordance to those studies which claim that macroalage may serve as good biomonitors of metal contamination in waters (Coelho et al., 2005; Kamala-Kannan et al., 2008; Wang and Dei, 1999).

A 10-fold increase in the initial concentration of Hg in solution resulted in a rise on the initial bioaccumulation rate of 3.6–4.6-fold (Table S3).

All kinetic models showed high performances in the description of results (Fig. 6 and Table S3), in most cases with better correlations ($0.975 < R^2 < 0.996$) than those recorded for Cd and Pb. Although values of q_t at 144 h, for 50 and 100 µg L^{-1}, were better estimated by PFO model (Fig. 6B–C), AIC suggests that PSO is the model more likely to be correct for all concentrations (higher R^2 and lower Sy/x). Thus, kinetic analysis suggests a Hg sorption of chemical nature, relying on chemical interactions between Hg ions and superficial functional groups of macroalgae surface (Bulgariu

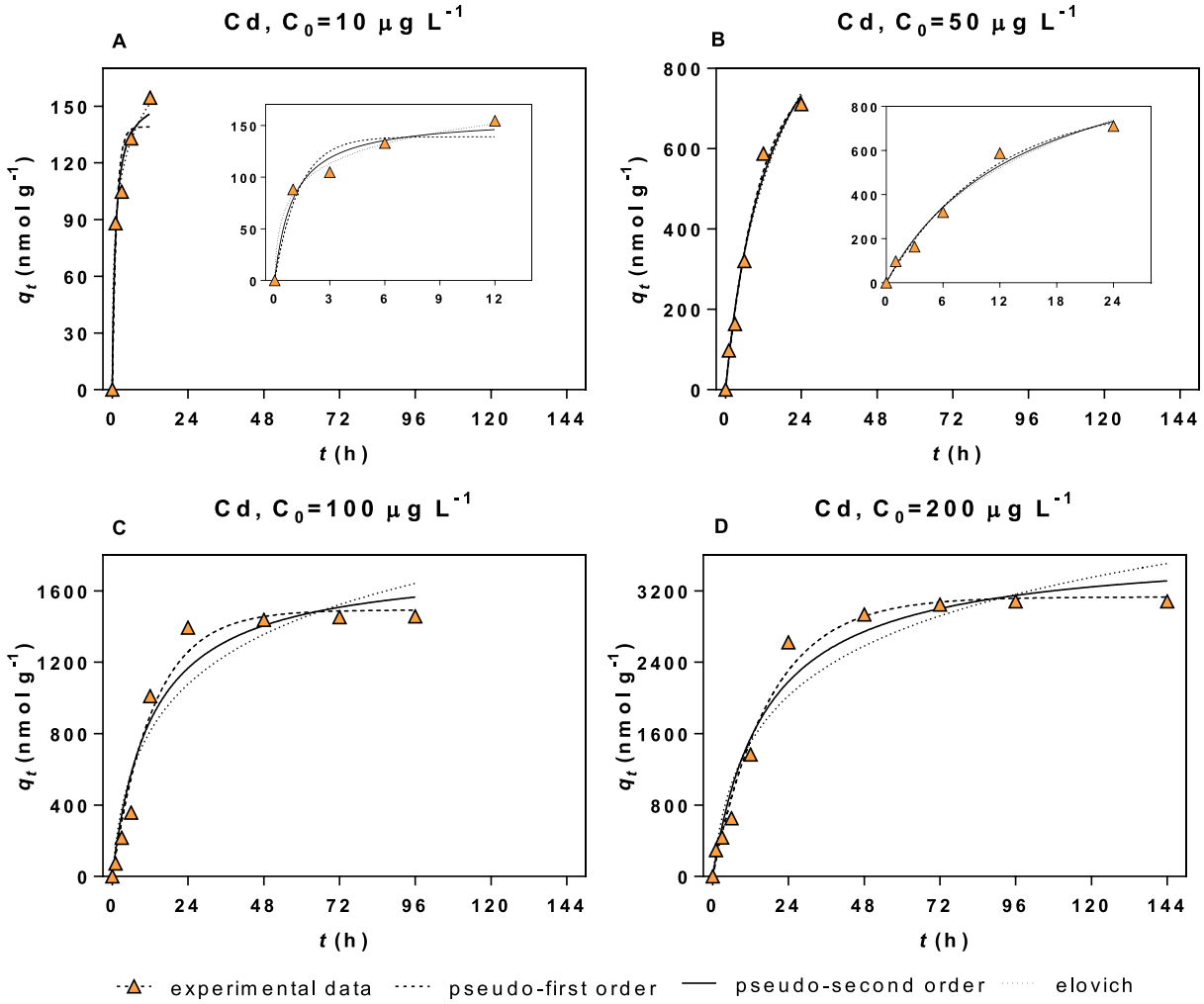


Fig. 2. Kinetic modelling of the experimental data regarding the uptake of Cd by living *U. lactuca* (mono-metallic solutions). Error bars were omitted for clarity.

and Bulgariu, 2014), such as ion-exchange and chemical adsorption. Along with those interactions, metabolism-dependent incorporation into cells should control bioaccumulation rates (Chojnacka, 2007; Vasconcelos and Leal, 2001).

3.2.1. *Metal concentration in algal biomass and method efficiency (mono-metallic solutions)*

Results in Table 2 show that, in percentage terms, green macroalgae have a similar and impressive ability to remove different metals from mono-metallic solutions, since in general removal were between 93% and 99%. However, comparing the removal efficiencies of *U. lactuca* toward the metals studied, in terms of time of contact required, it is possible to conclude that Cd removal is more efficient, since in most cases, it was more rapid (Figs. 1, 3 and 5). The content of Cd in macroalgae, measured after tissue acid digestion, for the highest initial concentration (C_0,Cd = 200 µg L^{-1}; $q_{t,144h.obtained}$ = 2876 ± 94.0 nmol g^{-1}) was very close to the value predicted ($q_{t,144h.predicted}$ = 3083 nmol g^{-1}) by mass balance (eq. (2)) between metal concentrations in solution at the beginning and end of the trial (relative error of 6%). This indicates that all Cd removed from the solution is actually bound in the seaweed biomass. Furthermore, chemical quantification of Cd in algal biomass, allowed to confirm the assumption that for concentrations of 10–100 µg L^{-1}, removal have continued beyond the point where it was possible to quantify this metal in solution, reaching virtually 100% of removal. The outstanding capability of *U. lactuca* to accumulate and retain Cd, gives it a high potential for application not only in processes of water treatment, but also in processes of pre-concentration for analytical chemistry (Romero-Gonzalez et al., 2000).

For Pb, results showed to be different, since only 46–58% of all metal removed from the solution showed to be in the macroalgae biomass, which suggests that other mechanisms may intervene in the remediation of water, such as precipitation (Chen and Wang, 2007). As concentrations remained stable in controls, precipitation of Pb had to be induced by the macroalgae, indirectly by raising the pH of solution (there was a gradual ascent along the experience until 9.3 ± 0.3), or directly through microprecipitation on alga surface (Chojnacka, 2009) and/or by production of exudates capable of complexing metals (Coelho et al., 2005; Torres et al., 2008). Further studies are required in order to understand those mechanisms.

Overall, the concentrations of Cd and Pb in macroalgae biomass were much higher than those reported by Masakorala et al. (2008) in their study about the influence of synthetic surfactants on the uptake of Cd and Pb by Ulva lactuca from mono-metallic solutions.

As seen for Cd, determination of Hg levels in macroalgae biomass at 144 h, confirmed that all metal removed from seawater along time was in the *U. lactuca* (Table 2). Thus, H_0 volatilization following Hg-reduction (Sunda and Huntsman, 1998) may be

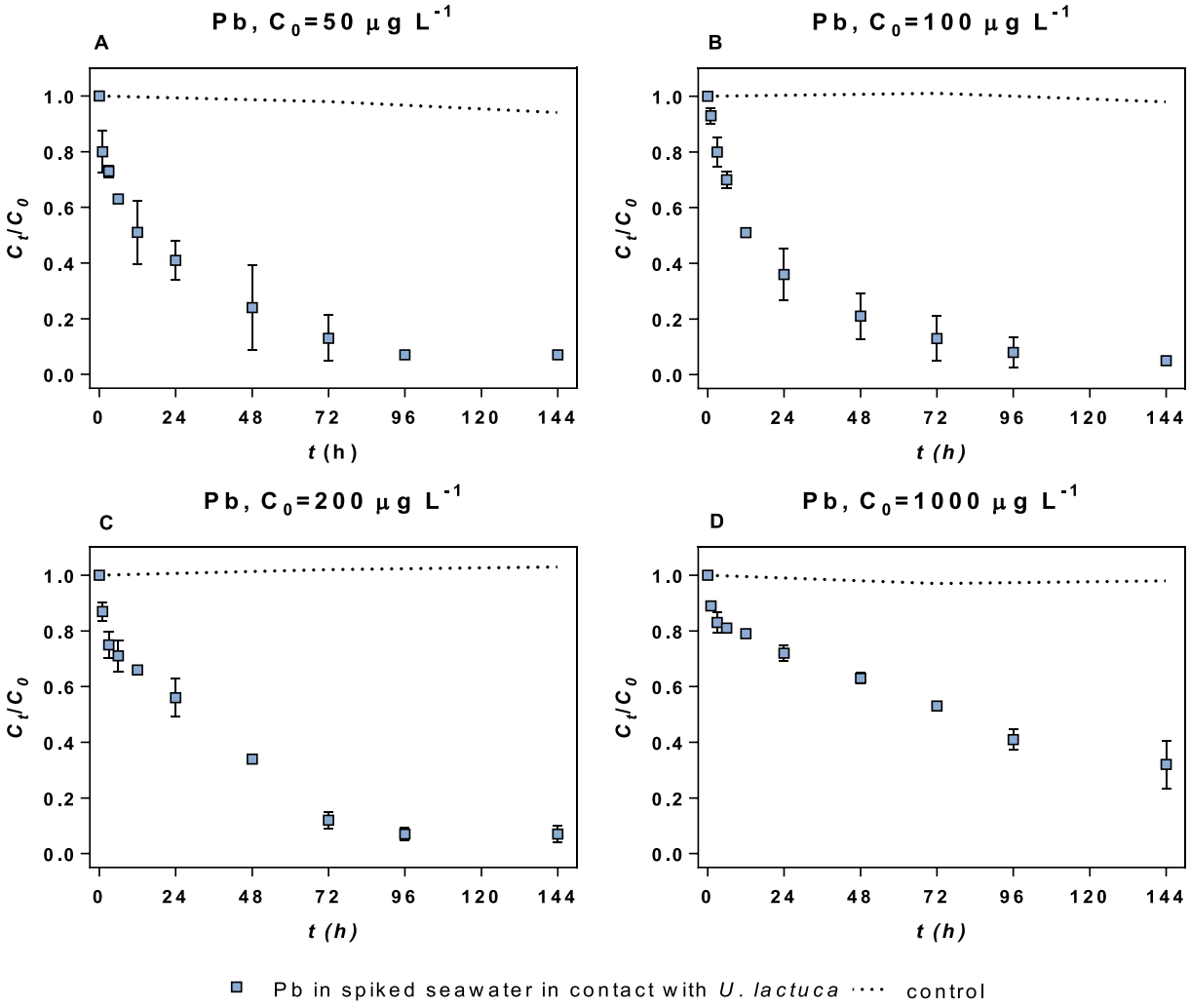


Fig. 3. Normalized concentrations (C_t/C_0) of Pb in spiked seawater along time (t, h) – mono-metallic solutions. Results are expressed as the mean ± standard deviation (n = 3).

excluded (Henriques et al., 2015).

Bioconcentration factors (BCF) in *U. lactuca*, defined as the ratio between metal concentrations in the organisms and the initial metal concentration in seawater, ranged between 1616 and 2169 for Cd, 960–1160 for Pb and 1855–2010 for Hg. Overall, BCF varied inversely with the concentration gradient in seawater (Coelho et al., 2005; Wang and Dei, 1999).

Overall, data demonstrate the effectiveness of *U. lactuca* in removing metals from salt water. Less than half-gram per litter of this green macroalgae allowed to reduce all concentrations of Hg in solution to values equal or below the limit set by the European Union for waters for human consumption, 1 µg L^{-1} (Council Directive 98/83/EC, 1998). Moreover, environmental quality standards (EQS) for Hg in surface water (Directive 2013/39/EU, 2013) were also achieved for the lowest initial concentration. Legal criteria were also met for Cd, since all residual levels of metal in solution were below the maximum permissible values (MPV) for drinking water, 5 µg L^{-1} (Decree-Law No. 103/2010, 2010), and bellow or very close to the maximum allowable concentration (MAC) of EQS for Cd in surface waters (Directive 2013/39/EU, 2013).

As for Pb and initial concentrations of 50 and 100 µg L^{-1}, levels of metal in solution at the end of the process were below 10 and 7.2 µg L^{-1}, which are respectively MAC values for safe drinking water, according to Portuguese law (Decree-Law No. 103/2010, 2010) and annual average (AA) of EQS for Pb in surface waters (Directive 2013/39/EU, 2013). For 1000 µg L^{-1}, final concentration of Pb in seawater exceeds the allowed values by about 40 times. However, it is important to note that for this concentration, no steady-state was observed, and if exposure time was prolonged, macroalgae would have improved water quality even more.

3.3. Water bioremediation in multi-metallic solutions

Results presented in Fig. 7A–B shows that *U. lactuca* is able to remove simultaneously Cd, Pb and Hg, yet with a removal efficiency for each metal depending greatly on the exposure condition. In M1, the profiles of curves C_t/C_0 were similar among metals, and removal percentages at the end of assays were also comparable, c.a. 96%, suggesting an equal affinity of green macroalgae toward different metals. However, it should be noted that initial molar concentration of Cd was ca. 1.8 folds the molar concentrations of Pb and Hg, and initial bioaccumulation rates varied among metals, according to Pb > Hg > Cd. This sequence is in agreement with the values of the kinetic constants of PFO (k_1) and PSO (k_2) models (Table S4), and follows an inverse trend of the initial metal concentrations ($C_{0,Cd} > C_{0,Hg} > C_{0,Pb}$, molar concentrations). The kinetic pattern observed in M2 was very different among the three metals, with levels of Hg being reduced very fast, reaching up to 99.7% of

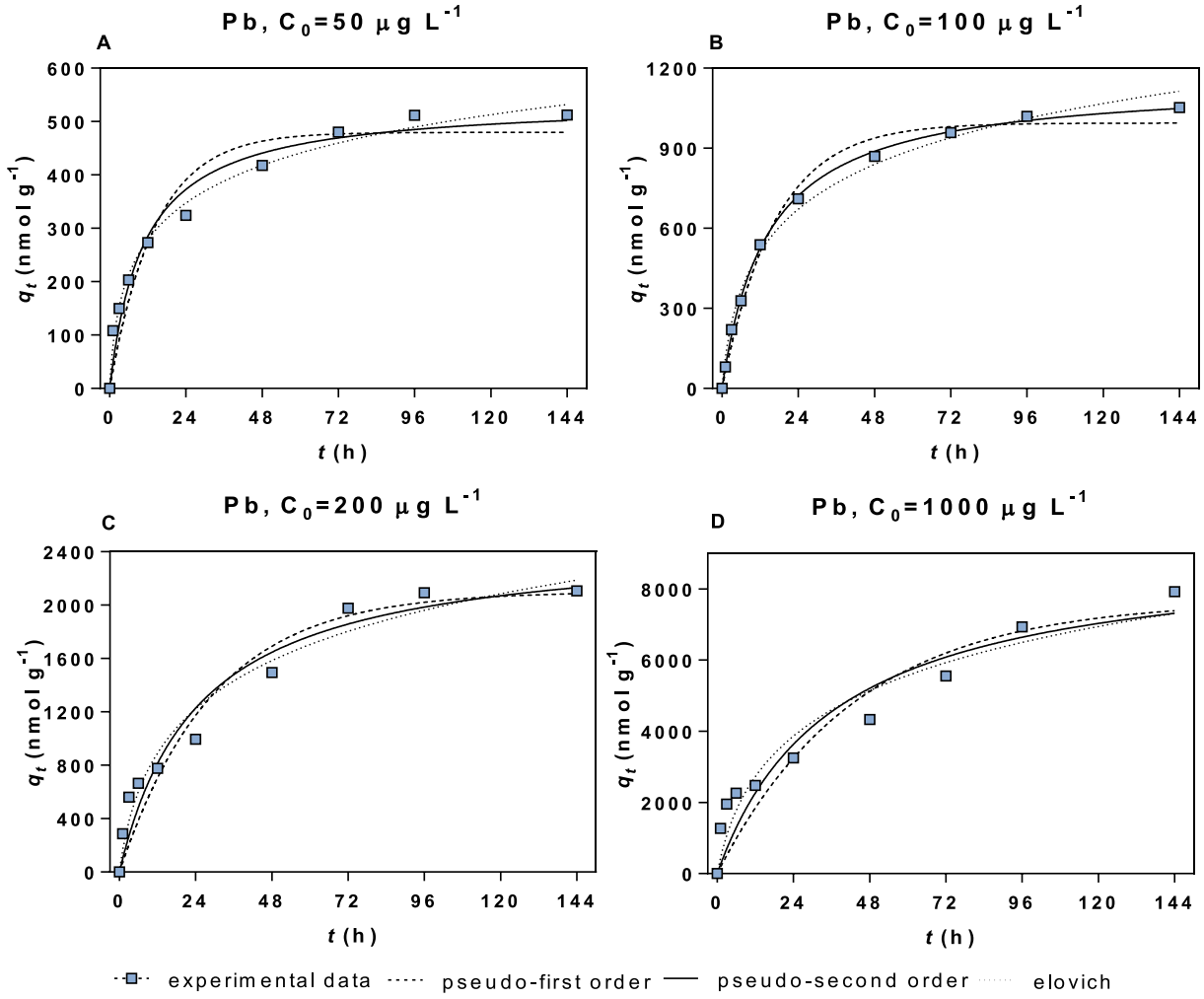


Fig. 4. Kinetic modelling of the experimental data regarding the uptake of Pb by living *U. lactuca* (mono-metallic solutions). Error bars were omitted for clarity.

removal at 144 h, while removal efficiencies for Cd and Pb were respectively 57.1 and 33.8%, with no equilibrium achieved. Although lower efficiencies were recorded for the metals with high initial concentration, an inverse situation was observed for the amount of metal incorporated by the macroalgae. *U. lactuca* exposed to M2 showed a total metal content of 6398 nmol g^{-1}, which is 3-fold that found for M1 (2060 nmol g^{-1}).

The selectivity of *U. lactuca* toward the different metals may be quantified by the ratio between the distribution coefficients, *K*, which in turn are given by $K = q_e/C_e$ for each metal at equilibrium (Lopes et al., 2014). Assuming the experimental values of $qt_{,144h}$, the pseudo-equilibrium selectivity found for the living macroalgae toward the target metals followed the order Cd > Hg > Pb in M1, whereas for M2 the order of selectivity was Hg > Cd > Pb. It seems that relative initial concentrations play a fundamental role in selectivity: for the low contamination scenario, selectivity follows the trend of increasing molar concentrations, while for a scenario of severe contamination the behavior is inverse. Kamala-Kannan et al. (2008) also found that the concentrations of metal in *U. lactuca* from Pulicat Lake (with levels of metal in water of 0.04–3.4 µg L^{-1}) decreased in the order: Cd > Pb, similar to that of surrounding environment.

Comparing the time evolution of normalized concentrations of each metal in mono- and multi-metallic solutions (Fig. 8A–F) it is possible to conclude that Hg was barely affect by the presence of Pb and Cd in both exposure conditions studied. However, after 48 h the reduction in Hg concentration was slightly slower at M1, compared to Hg removal rate in the absence of competing metals, or at M2 (Fig. 8E–F). The amount of Hg bound per unit of macroalgae mass was very similar between M1 and M2, respectively, 539 and 536 nmol g^{-1}.

Unlike what was observed for Hg, the profile of Cd removal was altered by the presence of competing metal ions, and differences were more noteworthy at M2. Initial bioaccumulation rates were lower than those recorded in the absence of Hg and Pb, and the extension of process was also negatively affected. After 144 h, living *U. lactuca* accumulated 1022 and 2323 nmol Cd g^{-1}, respectively for M1 and M2, being a longer time required to achieve an equilibrium in M2.

In the case of Pb, the kinetics of removal was affected differently depending on the contamination scenario. In M1, as verified for Hg, the removal of Pb was initially very fast, followed by a subtle decrease in C_t/C_0, until a balance is established, which corresponds to a bioaccumulation of 498 nmol g^{-1}. This removal pattern was very similar to that in the absence of competing metal ions in solution. In the exposure condition M2, a change in the kinetic profile due to the presence of other metals was evident, with a less pronounced decrease on Pb content in seawater, over time. The removal process was far from reaching equilibrium at 144 h, although the algae had already incorporated an impressive amount

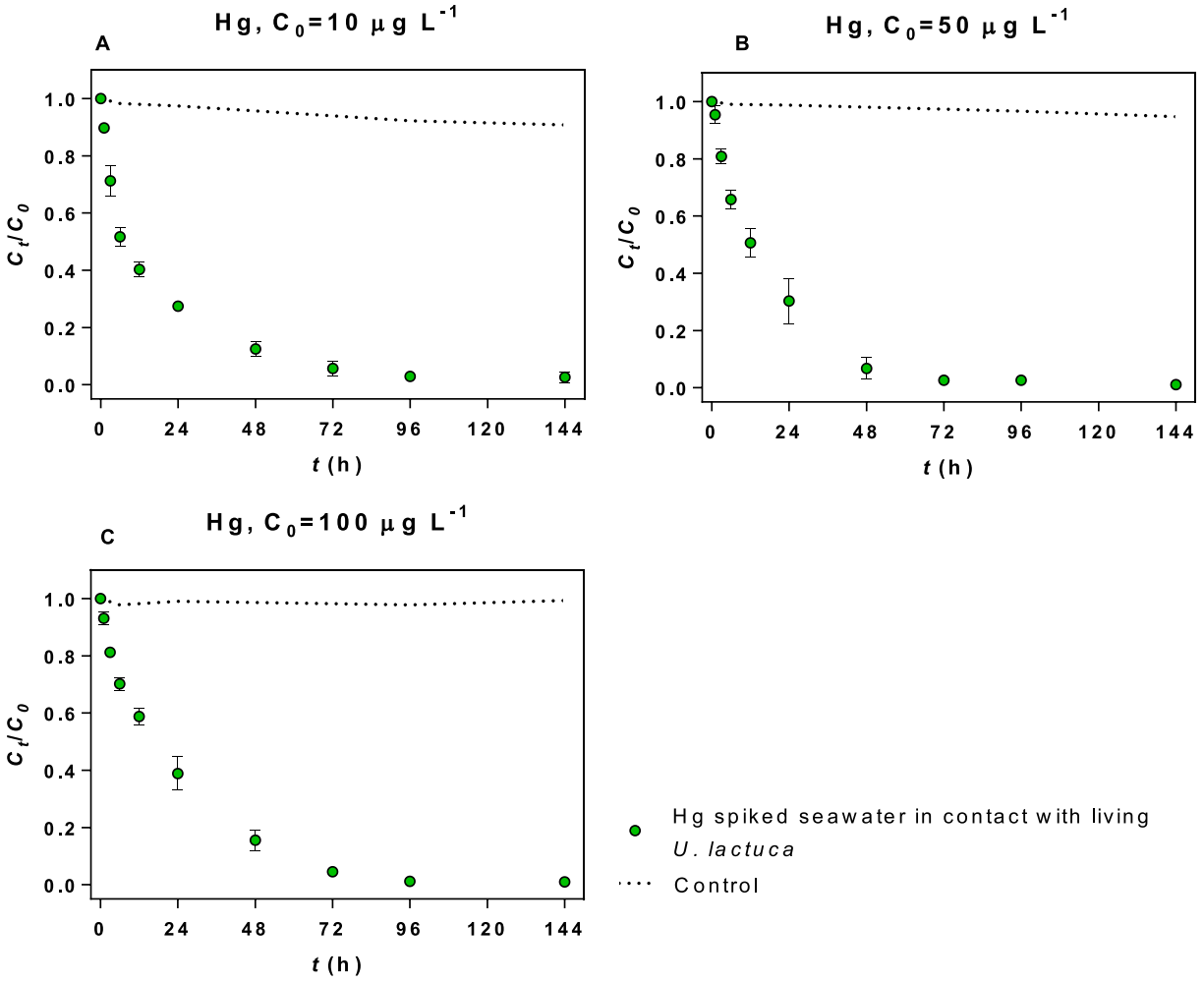


Fig. 5. Normalized concentrations (C_t/C_0) of Hg in spiked seawater along time (t, h) – mono-metallic solutions. Results are expressed as the mean ± standard deviation (n = 3).

of Pb, 3539 nmol g^{-1}.

The comparison between q_t values at equilibrium for a given metal, obtained in multi- and mono-metallic solutions, enable to assess the effect of the presence of foreign metals on the bioaccumulation. If $q_e^{multi}/q_e^{mono}<1$, the bioaccumulation is inhibited by the presence other metals; if $q_e^{multi}/q_e^{mono}=1$, there is no observable interaction; and if $q_e^{multi}/q_e^{mono}>1$, the bioaccumulation is favoured by the presence of co-existing metals. For Hg, values found were 1.05 and 1.04, respectively for M1 and M2, which indicates that bioaccumulation of Hg was neither favoured nor inhibited in both systems. The bioaccumulation of Pb was not affected by the presence of Hg and Cd in M1 (1.03), whereas for M2 it was highly inhibited (0.45). As for Cd, bioaccumulation was slightly inhibited in the case of low contamination scenario (0.94), and somehow inhibited by the presence of Hg and Pb, in the severe contamination scenario (0.75).

Overall, there was a good agreement between experimental data obtained in M1 and applied models ($0.968 < R^2 < 0.998$) – Fig. S1 and Table S4, in supplementary material. For each metal, the best fit model is consistent with the observations in mono-metallic solutions: Cd (PFO); Pb (Elovich); Hg (PSO). In M2, the fits showed a worst performance, with exception of Hg ($0.973 < R^2 < 0.993$). Pseudo-first order and PSO models failed to describe the bioaccumulation of Cd (ambiguous), while for Pb coefficients of correlation ranged between 0.896 and 0.965.

3.3.1. Metal concentration in algal biomass and method efficiency (multi-metallic solutions)

A summary of the data acquired at the end of bioaccumulation trials, conducted in saline water, for multi-metallic solutions of Cd, Pb and Hg, M1 and M2, in contact with living *U. lactuca* is presented in Table 3.

The measurement of metal content in macroalgal biomass revealed, once again, that all Cd and Hg removed from the seawater was fixed in the organism, since obtained values of $q_{t,144h}$ matched perfectly with predicted values of $q_{t,144h}$ (1.4%<E_r<5.7%). By contrary, only 49% and 71% of all Pb removed from the water, in M1 and M2 respectively, showed to be biosorbed/bioaccumulated by the marine macroalgae. Results follow the trend observed in the experiments of exposure to mono-metallic solution, and are in accordance with the findings of Turner et al. (2008b), who conducted short-term experiments in synthetic seawater, in order to study the effect of humic substances and salinity in the uptake of Pd, Cd, Hg and Pb by *U. lactuca*. Researchers found that Hg fraction internalized was c.a. 100% and Pb fraction internalized was below 30% (Turner et al., 2008b).

Bioconcentration factors in M1 were also in line with those obtained in mono-metallic solutions, respectively, 2266, 1032 and 2078 for Cd, Pb and Hg, whereas for M2, BCF for Cd and Pb were slightly lower, respectively 1384 and 522, and similar for Hg, 2106.

Analysing the efficiency of the removal process in terms of

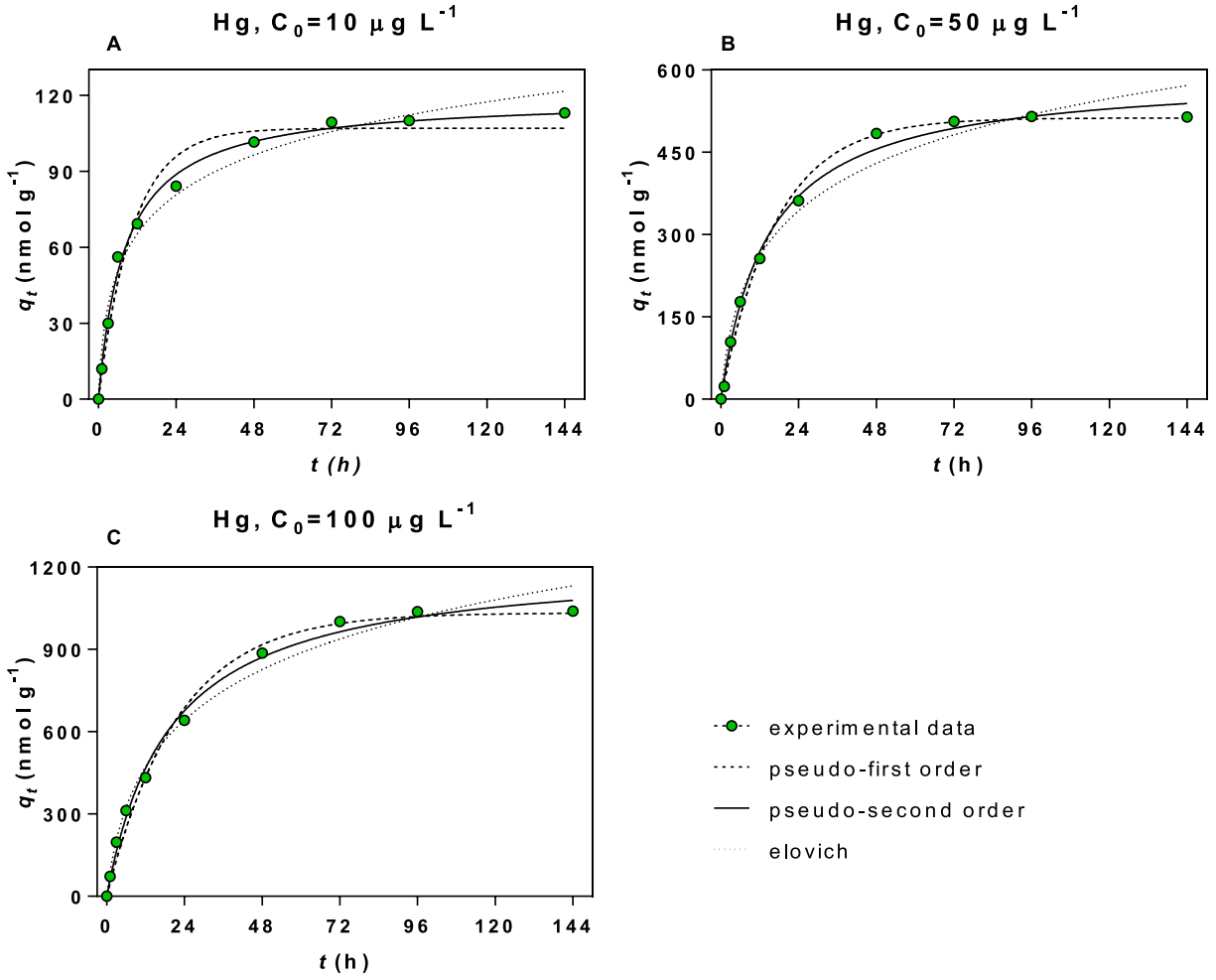


Fig. 6. Kinetic modelling of the experimental data regarding the uptake of Hg by living *U. lactuca* (mono-metallic). Error bars were omitted for clarity.

Table 2
Effect of initial metal concentrations on the amount of metal biosorbed by the *U. lactuca* ($q_{t,144h}$ – predicted and obtained), removal percentage and residual concentration of metal in solution ($C_{t,\ 144h}$).

Metal	C_0, µg L^{-1} (nmol L^{-1})	$q_{t,144h}$, predicted[a] nmol g^{-1}	$q_{t,144h}$, obtained[b] nmol g^{-1}	Removal$_{144h}$,[c] %	$C_{t,\ 144h}$, µg L^{-1}
Cd	10 (89)	>155	193 ± 15.8	>78.9 (100)	<2.0
	50 (445)	>711	822 ± 120	>94.9 (100)	<2.0
	100 (890)	>1457	1464 ± 136	>97.5 (98)	<2.0
	200 (1779)	3083	2876 ± 94.0	98.5	2.4
Pb	50 (241)	512	266 ± 79.2	92.6	4.0
	100 (483)	1053	561 ± 132	95.4	4.9
	200 (965)	2106	977 ± 130	93.3	16
	1000 (4826)	7922	4631 ± 35.8	67.7	379
Hg	10 (50)	113	93.2 ± 5.68	97.4	0.31
	50 (249)	514	501 ± 81.8	98.9	0.84
	100 (499)	1039	986 ± 101	99.0	1.0

[a] Through mass balance using eq. (2).
[b] Obtained by chemical quantification in seaweed.
[c] Obtained by using eq. (3). In brackets removal percentage obtained considering the chemical quantification of Cd in seaweed.

compliance with legal criteria on water quality, we can conclude that, for the lowest contamination scenario, MPVs for water for human consumption were achieved for Cd and Pb (Decree-Law No. 103/2010, 2010), as well as EQS for Pb in surface waters (Directive 2013/39/EU, 2013). The residual concentration of Hg in M1 was only slightly above the legal limits for drinking waters, however, in M2 the level of Hg was much lower than that allowable limit, due to synergistic effect of Cd and Pb. None of the legal criteria was accomplished for Cd and Pb in the highest contamination scenario, however it is important to remember that time profiles (Fig. 8B and D) indicate that better water quality would be attained if the period of contact time was extended. Moreover, it should be noted that the amount of algae used per volume of solution (1,6 g L^{-1}, fresh weight) was selected based on a previous work (Henriques et al., 2015), taking into account the minimization of the final residue. It is expected that an increase on the initial algal dosage would conduct to higher metal removal.

Based on all results, it was concluded that living *U. lactuca* may effectively be the basis of an alternative and cost-effective biotechnology for the simultaneous removal of Cd, Hg and Pb from contaminated waters.

3.4. Effects of metals on macroalgae growth

The presence of potentially toxic elements in the culture medium may affect organisms growth rate, and eventually lead to death, either by direct toxicity (Costa et al., 2011), or by nutrient deficiency (Sunda and Huntsman, 1998). The macroalgal growth rate is closely linked to the removal efficiency, since a fast growth will lead to a rapid increase of biomass, and consequently to a multiplication of specific sites for metal binding (Chojnacka, 2010).

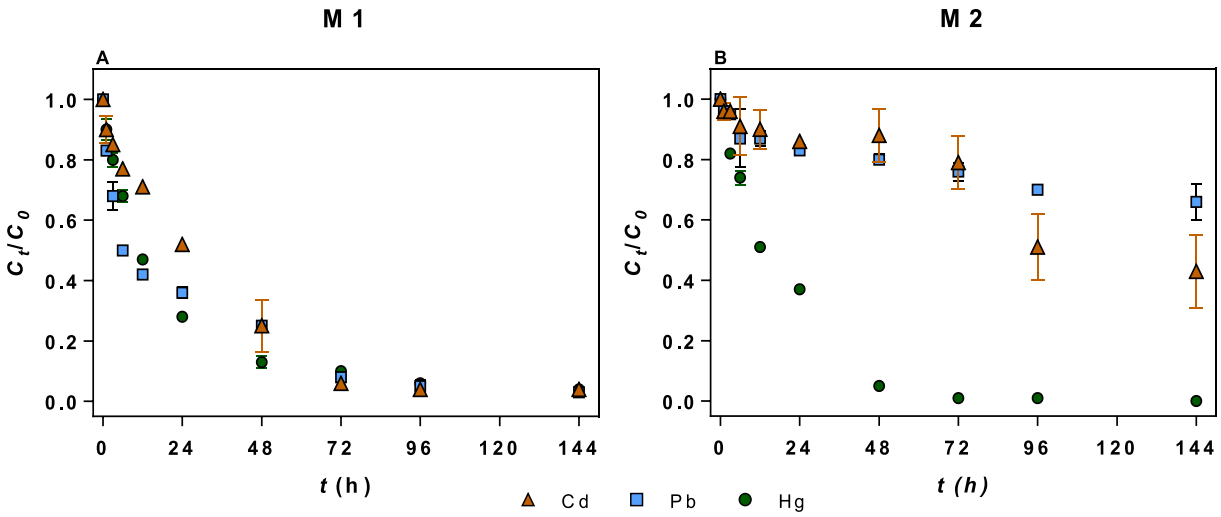


Fig. 7. Normalized concentrations (C_t/C_0) of Cd, Pb and Hg, simultaneously present in seawater along time (t, h), when in contact with living *U. lactuca*, under two different contamination scenarios, M1 and M2. The results are expressed as the mean ± standard deviation (n = 3).

Table 3
Summary of results obtained for multi-metallic solutions, M1 and M2, in contact with living *U. lactuca*: initial bioaccumulation rate (v_0) and amount of metal bioaccumulated by the *U. lactuca* ($q_{t,144h}$ – theoretical and experimental), removal percentage and residual concentration of each metal in solution ($C_{t,\ 144h}$) at the end of trials.

Mixture	Metal: C_0, nmol L^{-1}	$q_{t,144h}$ predicted[a] nmol g^{-1}	$q_{t,144h}$ obtained[b] nmol g^{-1}	Rem.,$_{144h}$[c] %	$C_{t,\ 144h}$ µg L^{-1}
M1	Cd: 445	1022	1008 ± 206.9	96.3	2.0
	Pb: 241	498.4	248.5 ± 12.17	96.7	2.0
	Hg: 249	539.4	518.4 ± 10.79	96.4	2.0
M2	Cd: 1779	2323	2463 ± 63.34	57.1	77.3
	Pb: 4826	3539	2519 ± 249.1	33.8	718
	Hg: 249	536.1	525.3 ± 12.37	99.7	0.15

[a] Through mass balance using eq. (2).
[b] Obtained by chemical quantification in seaweed.
[c] Obtained through variation of metal concentration in solution using eq. (3).

By other hand, macroalgae decay during the removal process would make unfeasible its application on water remediation, since metal accumulated would be released back into the water. Thus, the influence of Cd, Pb and Hg in *Ulva lactuca* growth rate was evaluated along the bioaccumulation experiments, for the mono- and multi-metallic solutions. It should be highlighted that for the full range of concentration used, no sporulation or marked loss of color, which is indicative of algae decay (Han and Choi, 2005) was observed during the 6 days of trials.

Fig. 9A–C shows the relative growth rate (RGR) of *Ulva lactuca*, expressed in % day^{-1}, at defined time periods of exposure to mono-metallic solutions. After 24 h of contact time, the RGR of reference (macroalgae in clean seawater) was 10.6 ± 3.3% day^{-1}, which is in line with the values recorded by Han et al. (2008) for two macroalgae of the genus *Ulva* (c.a. 12%).

On the whole, different growth patterns were observed between reference and the various contaminated solutions, however, with significant differences (two-way ANOVA, Tukey post hoc test, $\rho < 0.05$) only for shorter exposure time (24 h).

Cadmium (Fig. 9A) seems to enhance the cellular metabolism of the organism, causing a significant increase of RGR, at 24 h, in 10 and 50 µg L^{-1} when compared with reference (Tukey test, Q = 4.172 and Q = 4.572, respectively; $\rho < 0.05$). The reasons for that unexpected faster growing are not clear, and may be a response of the macroalgae against high levels of metal in cells, since increasing its biomass a "dilution effect" occurs (Costa et al., 2011; Sunda and Huntsman, 1998).

With the increase of contact time, differences between RGR of reference and Cd spiked seawaters become smaller, with almost none differences at 144 h. A similar behavior has been reported by Costa et al. (2011) when exposed *U. lactuca* to different concentrations of Hg in seawater. The authors explained this behavior by the fact that the metal concentration in solution decreases over time, but also by a possible physiological adaptation of the macroalgae to the presence of metal.

On the other hand, there was a trend towards decrease of *Ulva lactuca* RGR along time, i.e., in all cases macroalgae grew (RGR is positive) but at a progressively slower rate. This pattern was also verified for Pb and Hg, and may be attributed to the increasingly lower availability of nutrients and CO_2, which were consumed as algae grew.

At 24 h (Fig. 9B), only the RGR associated to Pb,C_0 = 50 µg L^{-1} was considerably higher (21.0 ± 7.3% day^{-1}) than RGR of reference (Tukey test, Q = 4.744; $\rho < 0.05$). Additionally, for the highest concentration of Pb, 1000 µg L^{-1}, a decrease on RGR was observed (8.2 ± 3.2% day^{-1}) when compared to the reference, however with no significance (two-way ANOVA, $\rho > 0.05$).

Exposure to Hg (Fig. 9C), affected growth rate positively ($RGR_{Hg}/RGR_{reference}>1$) but two-way ANOVA test only indicated significant differences between reference and 50 µg L^{-1}. As observed for Cd and Pb, RGRs of macroalgae exposed to Hg decreased over time for all concentrations, which can be explained by the reduced availability of nutrients and CO_2 in solution, capable of sustain growth rates similar to those observed in the beginning. Thus, for longer times of exposure it is very important to renew the solution or to add nutrients to the solution.

Along the experimental trials, RGRs of macroalgae in contact with multi-metallic solutions (Fig. 10) were in most cases higher than the correspondent RGR of reference. Significant differences were only observed at 24 h, between control and M1 (Tukey test, Q = 4.175; $\rho < 0.05$), which suggests that, for the concentrations tested, growth is not adversely influenced by the simultaneous presence of metals in solution. In general, the growth pattern was very similar for the two contamination scenarios, with exception of RGRs at 24 h, where M1 RGR (18.4 ± 6.1% day^{-1}) was significantly higher than that of M2 (10.7 ± 3.6% day^{-1}). Once again, macroalgae

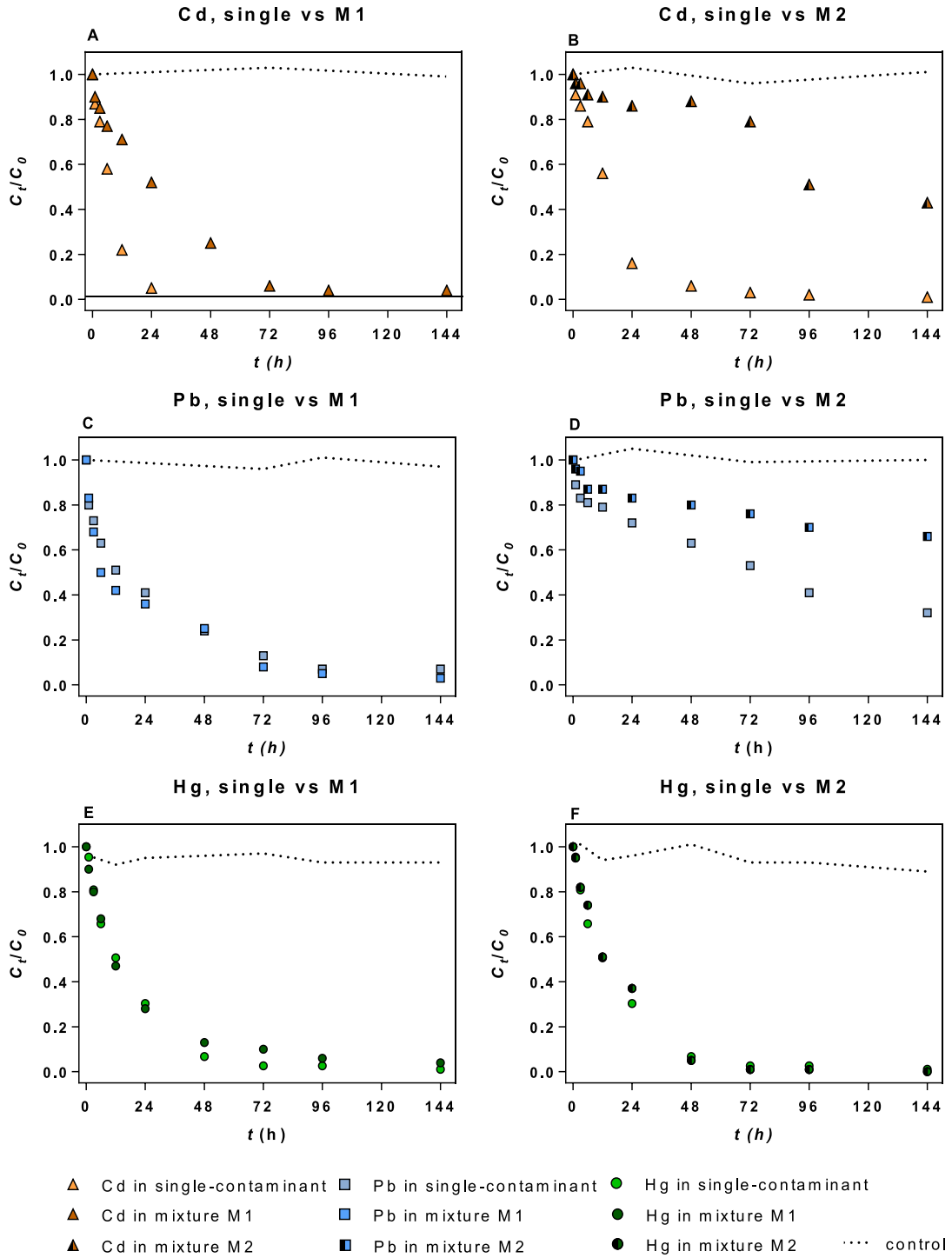


Fig. 8. Normalized concentrations (C_t/C_0) of Cd, Pb and Hg in spiked seawater along time (t, h), when in contact with *U. lactuca* (mono *vs.* multi-metallic solutions). Error bars were omitted for clarity.

RGRs decreased with time in all cases.

4. Conclusions

The effectiveness of living *U. lactuca* in the removal of metals from saline water was confirmed under different contamination scenarios. Fast-growing marine macroalgae were able to bioconcentrate Pb, Cd and Hg in their tissues 500–2200-fold the initial concentrations of solution, leading to high metal removal percentages. Initial bioaccumulation rates, and final amount of metal bound in macroalgae biomass were positively correlated with initial concentrations of metal in solution.

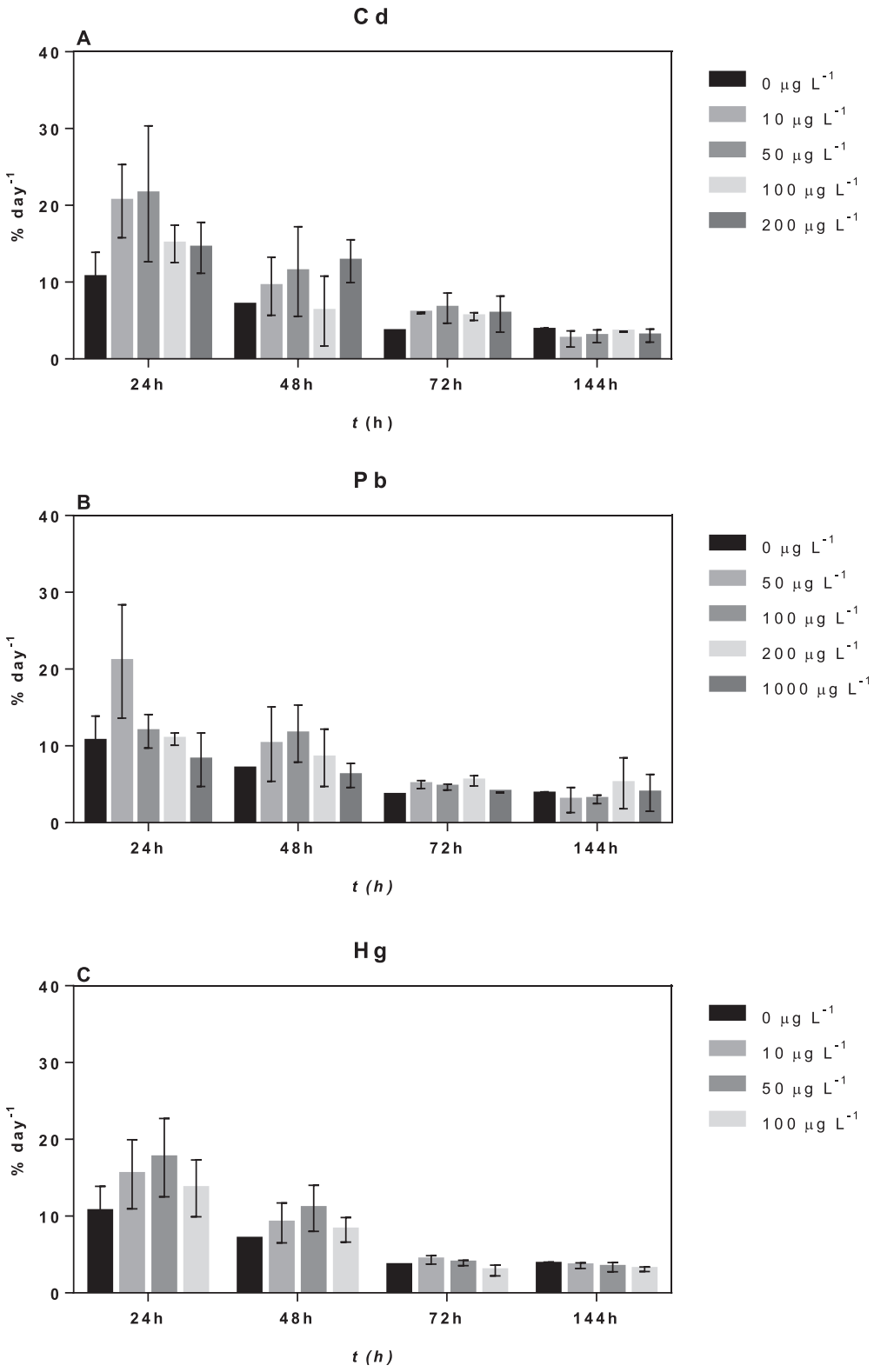


Fig. 9. Relative growth rate of *Ulva lactuca* in clean seawater (0 µg L^{-1}), and exposed to mono-metallic solutions of Cd (A), Pb (B) and Hg (C), in different concentrations, during 24, 48, 72 and 144 h. Results are expressed as mean and errors bars represent standard deviations (n = 6).

For mono-metallic solutions of Cd and Hg, removal efficiencies were always above 97%, representing final concentrations of metal in solution below, or very close to both surface and drinking water legal limits.

Under metal competition (ternary solutions), living *U. lactuca* was able to simultaneously remove Pb, Cd and Hg, although removal efficiencies were dependent of exposure condition. For the lowest contamination scenario, all removal percentages were

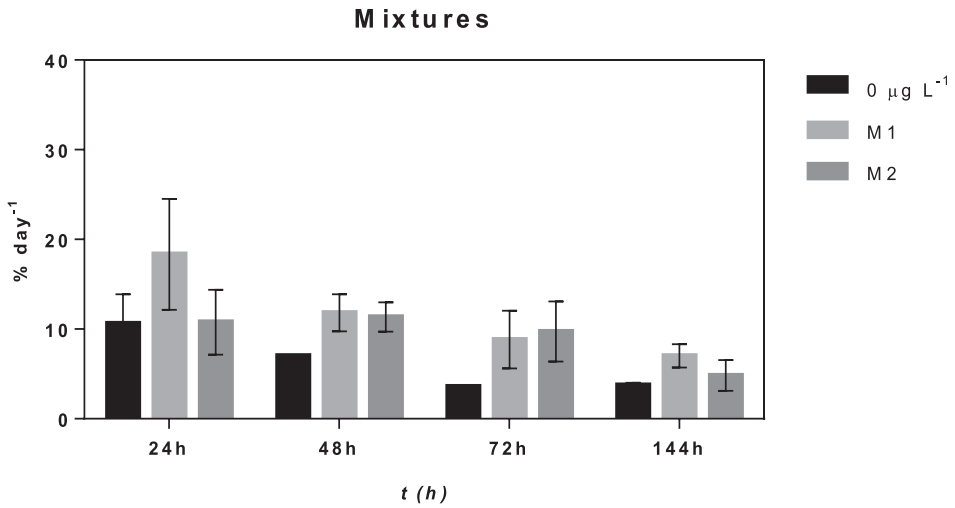


Fig. 10. Relative growth rate of *Ulva lactuca* in clean seawater (0 µg L^{-1}), and in contact with spiked multi-metallic solutions of Cd, Pb and Hg in equal mass concentrations (M1) and in different concentrations of the three elements (M2).

above 96%, whereas for the highest contamination scenario, removal efficiencies were c.a. 99% for Hg, 57% for Cd and 33% for Pb. Thus, increasing the level of contamination, a synergistic effect was observed for Hg, while the presence of other metals affected the removal of Cd and Pb. Under those conditions *U. lactuca* showed significant different affinities toward studied metals: Hg > Cd > Pb.

The bioaccumulation kinetic data was well described by the PFO, PSO and Elovich models, although in most cases the best fit model was different for each metal-macroalgae system.

Metal quantification in macroalgae biomass, after bioaccumulation, demonstrated that all Cd and Hg removed from the aqueous solution was strongly bound by the macroalgae, while only c.a. 50% of Pb proved to be in the macroalgae. This suggests that another mechanism, such as precipitation, occurred concurrently, leading to decreased levels of this metal in the solution over time.

It is noteworthy that in none of bioaccumulation assays occurred death of the organism. As regards to seaweed growth rate at the end of the tests, and for the full range of concentration studied, it was concluded that RGR was not negatively affected by the presence of Cd, Pb and Hg, even under multi-stressor exposure.

Overall, results highlighted the remarkable affinity of living *U. lactuca* to metals, which gives it a huge potential application in the treatment of saline waters.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jenvman.2017.01.035.

References

Bruhn, A., Dahl, J., Nielsen, H.B., Nikolaisen, L., Rasmussen, M.B., Markager, S., Olesen, B., Arias, C., Jensen, P.D., 2011. Bioenergy potential of Ulva lactuca: biomass yield, methane production and combustion. Bioresour. Technol. 102, 2595–2604.

Bulgariu, L., Bulgariu, D., 2014. Enhancing biosorption characteristics of marine green algae (Ulva lactuca) for heavy metals removal by alkaline treatment. Bioprocess. Biotech. 4.

Caliceti, M., Argese, E., Sfriso, A., Pavoni, B., 2002. Heavy metal contamination in the seaweeds of the Venice lagoon. Chemosphere 47, 443–454.

Chen, C., Wang, J., 2007. Influence of metal ionic characteristics on their biosorption capacity by Saccharomyces cerevisiae. Appl. Microbiol. Biot. 74, 911–917.

Cheung, C.W., Porter, J.F., McKay, G., 2000. Sorption kinetics for the removal of copper and zinc from effluents using bone char. Sep. Purif. Technol. 19, 55–64.

Chojnacka, K., 2007. Bioaccumulation of Cr(III) ions by blue-green alga *Spirulina sp.* Part I. A comparison with biosorption. Am. J. Agric. Biol. Sci. 2, 218–223.

Chojnacka, K., 2009. Biosorption and Bioaccumulation in Practice. Nova Science Publishers, Inc., New York.

Chojnacka, K., 2010. Biosorption and bioaccumulation - the prospects for practical applications. Environ. Int. 36, 299–307.

Coelho, J.P., Pereira, M.E., Duarte, A., Pardal, M.A., 2005. Macroalgae response to a mercury contamination gradient in a temperate coastal lagoon (Ria de Aveiro, Portugal). Estuar. Coast. Shelf Sci. 65, 492–500.

Costa, S., Crespo, D., Henriques, B.M.G., Pereira, E., Duarte, A.C., Pardal, M.A., 2011. Kinetics of mercury accumulation and its effects on Ulva lactuca growth rate at two salinities and exposure conditions. Water Air Soil Poll. 217, 689–699.

Council Directive 98/83/EC, 1998. Council Directive 98/83/EC of 3 November 1998 on the quality of water intended for human consumption. J. Eur. Communities 32–54.

Decree-Law No. 103/2010, 2010. Decree-law No. 103/2010 of the Portuguese Ministry of Environment and Spatial Planning of 24 September on Pollution Reduction. Diário da República, pp. 4289–4298. I Série.

Decree-Law No. 236/98, 1998. Decree-law No. 236/98 of the Portuguese Ministry of the Environment of 1 August Establishing Water Quality Standards. Diário da República, pp. 3676–3722. I Série.

Directive 2013/39/EU, 2013. Directive 2013/39/EU of the European Parliament and of the Council of 12 August 2013 amending Directives 2000/60/EC and 2008/105/EC as regards priority substances in the field of water policy. Official J. Eur. Union 1–17.

Directive 83/513/EEC, 1983. Council Directive 83/513/EEC of 26 September 1983 on limit values and quality objectives for cadmium discharges (Daughter to 2006/11/EC). Official J. Eur. Communities 1–8.

Directive 84/156/EEC, 1984. Council Directive 84/156/EEC of 8 March 1984 on limit values and quality objectives for mercury discharges by sectors other than the chlor-alkali electrolysis industry. Official J. Eur. Communities 49–54.

Eisler, R., 2010. Chapter 2-Algae and Macrophytes, Compendium of Trace Metals and Marine Biota. Elsevier, Amsterdam, pp. 7–97.

El-Khaiary, M.I., Malash, G.F., 2011. Common data analysis errors in batch adsorption studies. Hydrometallurgy 105, 314–320.

Farooq, U., Kozinski, J.A., Khan, M.A., Athar, M., 2010. Biosorption of heavy metal ions using wheat based biosorbents - a review of the recent literature. Bioresour. Technol. 101, 5043–5053.

Freitas, O.M.M., Martins, R.J.E., Delerue-Matos, C.M., Boaventura, R.A.R., 2008. Removal of Cd(II), Zn(II) and Pb(II) from aqueous solutions by brown marine

macro algae: kinetic modelling. J. Hazard Mater. 153, 493–501.
Han, T., Choi, G.-W., 2005. A novel marine algal toxicity bioassay based on sporulation inhibition in the green macroalga Ulva pertusa (Chlorophyta). Aquat. Toxicol. 75, 202–212.
Han, T., Kang, S.-H., Park, J.-S., Lee, H.-K., Brown, M.T., 2008. Physiological responses of Ulva pertusa and U. armoricana to copper exposure. Aquat. Toxicol. 86, 176–184.
He, J., Chen, J.P., 2014. A comprehensive review on biosorption of heavy metals by algal biomass: materials, performances, chemistry, and modeling simulation tools. Bioresour. Technol. 160, 67–78.
Henriques, B., Rocha, L.S., Lopes, C.B., Figueira, P., Monteiro, R.J.R., Duarte, A.C., Pardal, M.A., Pereira, E., 2015. Study on bioaccumulation and biosorption of mercury by living marine macroalgae: prospecting for a new remediation biotechnology applied to saline waters. Chem. Eng. J. 281, 759–770.
Ho, Y.S., Ng, J.C.Y., Mckay, G., 2000. Kinetics of pollutant sorption by biosorbents: Review. Separ Purif. Method 29, 189–232.
Jayakumar, R., Rajasimman, M., Karthikeyan, C., 2014. Sorption of hexavalent chromium from aqueous solution using marine green algae Halimeda gracilis: optimization, equilibrium, kinetic, thermodynamic and desorption studies. J. Environ. Chem. Eng. 2, 1261–1274.
Kamala-Kannan, S., Prabhu Dass Batvari, B., Lee, K.J., Kannan, N., Krishnamoorthy, R., Shanthi, K., Jayaprakash, M., 2008. Assessment of heavy metals (Cd, Cr and Pb) in water, sediment and seaweed (Ulva lactuca) in the Pulicat Lake, South East India. Chemosphere 71, 1233–1240.
Kumar, P., Govindaraju, M., Senthamilselvi, S., Premkumar, K., 2013. Photocatalytic degradation of methyl orange dye using silver (Ag) nanoparticles synthesized from Ulva lactuca. Colloids Surfaces B Biointerfaces 103, 658–661.
Lamare, M.D., Wing, S.R., 2001. Calorific content of New Zealand marine macrophytes. N. Z. J. Mar. Freshw. Res. 35, 335–341.
Lee, W.-Y., Wang, W.-X., 2001. Metal accumulation in the green macroalga Ulva fasciata: effects of nitrate, ammonium and phosphate. Sci. Total Environ. 278, 11–22.
Li, C., Xu, Y., Jiang, W., Dong, X., Wang, D., Liu, B., 2013. Effect of NaCl on the heavy metal tolerance and bioaccumulation of Zygosaccharomyces rouxii and Saccharomyces cerevisiae. Bioresour. Technol. 143, 46–52.
Lopes, C.B., Oliveira, J.R., Rocha, L.S., Tavares, D.S., Silva, C.M., Silva, S.P., Hartog, N., Duarte, A.C., Pereira, E., 2014. Cork stoppers as an effective sorbent for water treatment: the removal of mercury at environmentally relevant concentrations and conditions. Environ. Sci. Pollut. Res. Int. 21, 2108–2121.
Masakorala, K., Turner, A., Brown, M.T., 2008. Influence of synthetic surfactants on the uptake of Pd, Cd and Pb by the marine macroalga, Ulva lactuca. Environ. Pollut. 156, 897–904.
Monterroso, P., Abreu, S.N., Pereira, E., Vale, C., Duarte, A.C., 2003. Estimation of Cu, Cd and Hg transported by plankton from a contaminated area (Ria de Aveiro). Acta Oecol. 24 (Suppl. 1), S351–S357.
Nielsen, M.M., Bruhn, A., Rasmussen, M.B., Olesen, B., Larsen, M.M., Moller, H.B., 2012. Cultivation of Ulva lactuca with manure for simultaneous bioremediation and biomass production. J. Appl. Phycol. 24, 449–458.
Nikolaisen, L.S., Jensen, P.D., 2013. 3-Biomass feedstocks: categorisation and preparation for combustion and gasification. In: Rosendahl, L. (Ed.), Biomass Combustion Science, Technology and Engineering. Woodhead Publishing, pp. 36–57.
Rocha, L., Lopes, C., Borges, J.A., Duarte, A.C., Pereira, E., 2013. Valuation of unmodified rice husk waste as an eco-friendly sorbent to remove mercury: a study using environmental realistic concentrations. Water, Air, & Soil Pollut. 224, 1–18.
Romera, E., Gonzalez, F., Ballester, A., Blazquez, M.L., Munoz, J.A., 2007. Comparative study of biosorption of heavy metals using different types of algae. Bioresour. Technol. 98, 3344–3353.
Romero-Gonzalez, M.E., Williams, C.J., Gardiner, P.H.E., 2000. The application of dealginated seaweed as a cation exchanger foron-line preconcentration and chemical speciation of trace metals. J. Anal. Atomic Spectrom. 15, 1009–1013.
Sari, A., Tuzen, M., 2008. Biosorption of Pb(II) and Cd(II) from aqueous solution using green alga (Ulva lactuca) biomass. J. Hazard Mater. 152, 302–308.
Sekabira, K., Origa, H.O., Basamba, T.A., Mutumba, G., Kakudidi, E., 2011. Application of algae in biomonitoring and phytoextraction of heavy metals contamination in urban stream water. Int. J. Environ. Sci. Te 8, 115–128.
Sunda, W.G., Huntsman, S.A., 1998. Processes regulating cellular metal accumulation and physiological effects: phytoplankton as model systems. Sci. Total Environ. 219, 165–181.
Torres, M.A., Barros, M.P., Campos, S.C.G., Pinto, E., Rajamani, S., Sayre, R.T., Colepicolo, P., 2008. Biochemical biomarkers in algae and marine pollution: a review. Ecotoxicol. Environ. Saf. 71, 1–15.
Trinelli, M.A., Areco, M.M., Afonso, M.D., 2013. Co-biosorption of copper and glyphosate by Ulva lactuca. Colloid Surf. B 105, 251–258.
Turner, A., Pedroso, S.S., Brown, M.T., 2008a. Influence of salinity and humic substances on the uptake of trace metals by the marine macroalga, Ulva lactuca: experimental observations and modelling using WHAM. Mar. Chem. 110, 176–184.
Turner, A., Pedroso, S.S., Brown, M.T., 2008b. Influence of salinity and humic substances on the uptake of trace metals by the marine macroalga, Ulva lactuca: experimental observations and modelling using WHAM. Mar. Chem. 110, 176–184.
Vasconcelos, M.T.S.D., Leal, M.F.C., 2001. Seasonal variability in the kinetics of Cu, Pb, Cd and Hg accumulation by macroalgae. Mar. Chem. 74, 65–85.
Wang, J.L., Chen, C., 2009. Biosorbents for heavy metals removal and their future. Biotechnol. Adv. 27, 195–226.
Wang, W.-X., Dei, R.C.H., 1999. Kinetic measurements of metal accumulation in two marine macroalgae. Mar. Biol. 135, 11–23.
Yamashita, M., Tomita-Yokotani, K., Hashimoto, H., Sawaki, N., Notoya, M., 2009. Sodium and potassium uptake of Ulva – application of marine macro-algae for space agriculture. Adv. Space Res. 43, 1220–1223.